SMART INDIA HACKATHON 2026

SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID – 26047**
- **Problem Statement Title- Patient Case-Taking Software: a voice-first, multilingual AI platform for patient history and medical-document digitization**
- **Theme- Healthcare & MedTech [confirm on SIH portal]**
- **PS Category- Software/Hardware → Software**
- **Team ID- [Fill from SIH portal]**
- **Team Name (Registered on portal) [Fill Team Name]**



[Team Name]
INNOVATE • BUILD • HEAL

IDEA TITLE

PatientCase_AI: Voice-First AI Case-Taking

SMART INDIA HACKATHON 2026

Letting every patient **tell their story in their own language**, so the doctor starts the consultation with a **verified, source-referenced history**.

WHAT ?
Our Solution

- Patient-facing AI history taking in English, Hindi and Marathi
- Voice or touch, every question answerable both ways
- Follow-ups that fit what the patient said
- Reads prescriptions and lab reports

WHY ?
Need / Problem

- 4,000 to 10,000 OPD patients a day; 2 to 5 minute consultations
- History is under-elicited; allergies get missed
- Paper records go unread
- No first-mile capture into ABDM

WHERE ?
Deployment Areas

- Government and tertiary hospital OPDs
- Registration and waiting areas
- AYUSH OPDs (Dashavidha history)
- District hospitals and health centres

WHO ?
Target Users

- Patients, including rural, elderly and low-literacy
- Registration staff and nurses
- Consulting doctors
- Triage staff and administrators

WHEN ?
Use Cases / Timing

- While the patient waits, before the visit
- When an answer signals an emergency
- At the consultation, to read the history in seconds
- On return visits, records already digitized

HOW ?
Process Flow

- Register, pick language, consent
- Voice or touch interview, adaptive follow-ups
- Upload documents; AI digitizes them
- Summary to doctor and HIS (FHIR)

WHICH ?
Technology & Approach

- Bhashini / AI4Bharat speech
- LLM follow-ups inside safety checks
- Tesseract OCR, rule-based extraction
- React, Node.js, PostgreSQL, FHIR

System Workflow

Patient Terminal / Phone
- ✓ Language and consent
- ✓ Speak or tap answers
- ✓ Audio prompts and icons

Audio and answers →

Interview Engine
- ✓ Adaptive follow-ups (LLM)
- ✓ Fixed safety questions
- ✓ Red-flag rules, instant alert

Files →

Document AI
- ✓ Quality check, OCR
- ✓ Classify, extract values
- ✓ Timeline, drug and lab checks

Structured data →

Doctor Workspace
- ✓ Summary with sources
- ✓ Accept / amend / reject
- ✓ FHIR to HIS and ABHA

Innovation & Uniqueness

Voice and Touch
Every question works by speaking or tapping.

Adaptive, Not Fixed
The next question follows the patient's own words.

The System Decides
Every AI question is checked; the doctor decides.

Records Included
Files are read and checked against the answers.

Built for India
Marathi, Hindi, English and AYUSH history mode.

Consent First
Granular, revocable, audited; no data stored in voice.

[Team Name]
INNOVATE • BUILD • HEAL

TECHNICAL APPROACH

PatientCase_AI: Voice-First AI Case-Taking Platform

SMART INDIA HACKATHON 2026

A **browser-based** patient interface talks to a **Node.js API**; **Bhashini** handles Indian-language speech, and **fixed rules plus a checked LLM** run the interview.

1 SYSTEM ARCHITECTURE

Capture simply. Check everything. Let the doctor decide.

PATIENT TERMINAL / PHONE

- React 19 + Vite
- Icon-driven UI, large touch targets
- Records the answer, 16 kHz WAV
- English, Hindi, Marathi

→

BHASHINI / AI4BHARAT

- ASR: speech to text
- TTS: spoken prompts
- Browser speech as fallback
- Audio and words never stored

→

API (Node.js + Express)

- Role gateway: patient, staff, doctor, admin
- Consent checked on every route
- Interview engine + red-flag rules
- Real-time alerts (SSE)

→

AI + RULES

- LLM (Gemini) writes follow-up questions
- System refuses diagnoses, advice, repeats
- Tesseract OCR, extraction, analysis
- Falls back to the fixed question tree

→

POSTGRESQL

- Answers, files, analysis
- Versioned summaries
- Append-only audit log
- Encrypted identifiers

→

HIS / ABDM

- FHIR bundles with retry
- ABHA create and link
- Consent-gated delivery
- Stand-ins in the prototype

2 TECHNOLOGY STACK

- **Frontend** — React 19, Vite 6, React Router; inline SVG charts
- **Backend** — Node.js, Express 5, JWT cookie sessions, optional TOTP
- **Database** — PostgreSQL 18; 12 versioned migrations
- **Speech** — Bhashini / AI4Bharat ASR and TTS
- **Document AI** — Tesseract OCR, rule-based extraction
- **LLM** — Gemini (or Claude) behind safety checks

3 PATIENT INTERVIEW FLOW

Reason first. Then follow the story. Safety questions always.

1. Reason for Visit
- Said or typed in own words
- Complaint and duration read

2. Emergency Screen
- Confirmed on screen and by audio
- Red flag, instant staff alert

3. Adaptive Follow-ups
- LLM writes each next question
- Up to 6, stops when enough

4. System Checks
- No diagnosis or advice
- Three languages; no repeats

5. Medicines, Allergies
- Always asked, always last
- Dual confirmation

6. Review and Submit
- Patient can correct answers
- Plain-language recap

4 DOCUMENT AND DOCTOR FLOW

From a photo of a report to a source-referenced summary.

1. Upload
- Photos or PDFs, many files
- Quality check, retake advice

2. Read
- OCR: English and Devanagari
- Classify the document type

3. Extract
- Medicines, doses, lab values
- Dates and sources kept

4. Cross-check
- Timeline, medicine reconciliation
- Interactions, allergies, labs

5. Summarize
- Draft with a source on each line
- "Not answered" never means "no"

6. Doctor Decides
- Accept, amend or reject
- Pushed to HIS as FHIR

5 USE CASES

Patient	Staff	Doctor
Register, consent	Register patients	Read summary
Speak or tap	See alerts, queue	Accept / amend

6 KEY HIGHLIGHTS

- **Safe by design** — No diagnosis; the system overrules the model
- **Multilingual voice** — English, Hindi, Marathi, with tap fallback
- **Consent and audit** — Per-purpose consent; every action logged

[Team Name]
INNOVATE • BUILD • HEAL

FEASIBILITY AND VIABILITY

A Practical, Scalable and Safe Solution for Hospital OPDs

SMART INDIA HACKATHON 2026

TECHNICAL FEASIBILITY

Built with proven technologies and practical resources

Working Prototype

Patient, staff, doctor and admin portals run end to end, with 244 server and 43 web automated tests passing.

Proven Speech Technology

Bhashini / AI4Bharat for Indian-language ASR and TTS, with the browser's speech as a built-in fallback.

Commodity Hardware

Runs in a browser on any terminal or phone; a single server and PostgreSQL serve one hospital.

Open, Standard Components

Tesseract OCR, FHIR bundles and ABDM-style flows; no proprietary lock-in.

Pilot Readiness

One department, supervised, with synthetic data first; protocol and checklists already drafted.

POTENTIAL CHALLENGES AND RISKS

Real-world constraints and limitations

Noisy Wards and Accents

Speech accuracy, especially Marathi, can drop in crowded rooms; no field benchmark has been run yet.

LLM Reliability

A model can write an unsuitable question or wording; clinical sign-off on sample interviews is still needed.

Low-Literacy Usability

Real first-time patients must be able to finish unaided; a usability test has not been run.

Integration and Cost

Real HIS and ABDM connections, hosting and model quotas (free tiers are tiny) need approval and budget.

Privacy and Compliance

Sending patient text to outside AI needs hospital approval, DPDP-aligned consent and a security review.

STRATEGIES FOR OVERCOMING CHALLENGES

Practical solutions for reliable and safe use

Voice and Touch Together

Every question can be tapped; repeated speech failures offer tapping and then alert staff.

The System Overrules the Model

Diagnosis, advice, repeats and missing Hindi or Marathi are refused; failures fall back to the fixed question tree.

Clinician in the Loop

Doctors accept, amend or reject; red-flag rules and sample AI interviews go to clinical review before use.

Phased Rollout

Stand-ins first, then real OTP, HIS and ABDM; hospital-hosted Bhashini keeps audio inside the hospital.

Privacy by Default

AI off by default, only de-identified facts sent, voice never stored, append-only audit log.

OVERALL VIABILITY

Technically practical. Safe by design. Ready to pilot.

The solution is **feasible today** as a supervised pilot in one OPD and can scale across departments. It uses proven components, keeps the doctor in control, and lists its open risks plainly, which makes it a **viable and responsible** path to better histories.

[Team Name]
INNOVATE • BUILD • HEAL

IMPACT AND BENEFITS

Better Histories. Faster Consultations. Healthier Communities.

SMART INDIA HACKATHON 2026

IMPACT ON TARGET AUDIENCE

Empowering every stakeholder in the OPD

Patients

Tell their story in their own language, unhurried, with no reading needed.

Doctors

Start with a verified, source-referenced history and spend the visit on examination and counselling.

Registration Staff and Nurses

Fewer repeated questions; instant alerts and help requests.

Hospital Administrators

Dashboards, audit trail and operations status in one place.

AYUSH Practitioners

Extended Dashavidha Pariksha captured without eating into consultation time.

SOCIAL BENEFITS

Care that includes everyone

- Marathi, Hindi and English, so language is not a barrier to being heard.
- Icons and audio prompts make it usable for elderly and low-literacy patients.
- Emergency signs reach staff at once instead of waiting in the queue.
- Fewer missed allergies and medicines supports safer care.

ECONOMIC BENEFITS

Cost-effective at hospital scale

- Doctor and nurse time moves from asking to deciding.
- Standard software and a browser; no kiosk hardware is required.
- Fewer duplicate investigations when old reports are read.
- Open components keep licence costs low.

ENVIRONMENTAL BENEFITS

Greener, paperless care

- Paper intake forms and photocopies are replaced by digital records.
- Digitized prior reports reduce repeat tests and extra visits.
- Only needed data moves; raw audio is not stored.
- One lightweight server per hospital, no special hardware.

OPERATIONAL BENEFITS

Real-time intelligence for efficient OPDs

- Red-flag alerts and a live queue by department.
- Standard-format summary on the doctor's screen at consultation.
- FHIR delivery to the HIS and ABHA link, with retry.
- Audit log and monitoring support compliance reviews.

OVERALL IMPACT

"From a rushed two-minute conversation to a complete, verified history before the doctor says hello: faster consultations, safer care and a record the patient helped write."

RESEARCH AND REFERENCES

Building on Proven Research and National Standards

Consultation Time Across Countries

Irving et al., BMJ Open 2017

"International variations in primary care physician consultation time: a systematic review of 67 countries"

- Compared average consultation length across 67 countries.
- Placed India's average at just over 2 minutes.
- Shows why history must be captured before the doctor sees the patient.

BMJ Open 2017

Bhashini and AI4Bharat Speech Models

MeitY Bhashini; AI4Bharat, IIT Madras

Open ASR, TTS and translation for Indian languages, including IndicTrans2 (Gala et al., 2023)

- National language platform named in the problem statement.
- Covers English, Hindi and Marathi speech.
- Our speech layer calls it through one swappable adapter.

Language AI

ABDM and HL7 FHIR

National Health Authority; HL7 International

Ayushman Bharat Digital Mission; FHIR R4 interoperability standard

- Defines ABHA IDs and consent-based health information exchange.
- FHIR is the exchange format we deliver to the HIS.
- Addresses the "first-mile" gap before the encounter.

Interoperability

Privacy and AI Ethics in Healthcare

Digital Personal Data Protection Act 2023; ICMR 2023

DPDP Act 2023; ICMR "Ethical Guidelines for Application of AI in Biomedical Research and Healthcare"

- Granular, revocable consent and purpose limits.
- Human oversight of AI output; no autonomous diagnosis.
- Shapes our consent, audit and de-identification design.

Compliance

Open-Source OCR for Documents

R. Smith, ICDAR 2007; Tesseract OCR project

"An Overview of the Tesseract OCR Engine"

- Open OCR engine with English and Devanagari models.
- Used to read printed prescriptions and lab reports.
- Rule-based extraction keeps every value traceable to its file.

Document AI

Structured Clinical History

SOCRATES pain framework; Dashavidha Pariksha

Standard history elements for allopathic OPDs, and the ten-fold examination of Ayurveda (Charaka Samhita)

- Guides which follow-up questions help a doctor most.
- Dashavidha Pariksha is captured in AYUSH mode.
- Gives clinicians a familiar structure to verify.

Clinical Method

References listed from the team's reading; verify titles, authors and years against the sources before final submission.

KEY TAKEAWAY

Our solution is built on national standards and published research in consultation time, Indian-language speech, health-data exchange, privacy and clinical history taking, and combines them into one safe, voice-first platform that keeps the doctor in control.